

Intergenerational Housing Misallocation and Fertility

Short model note

Tommaso De Santo

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This note sets up a two-period model in which children require space, family-sized space sits behind a down payment, and the existing stock is held by older households whom the tax code pays to keep it. It defines the environment, solves the household problems, and derives the three objects the project runs on: the effective family-housing cap that prices children, the misallocation surplus between constrained buyers and locked-in incumbents, and the fixed-price fertility effect of housing policy. A longer companion document contains the proofs and the full equilibrium treatment.

1 Environment

The economy is populated by overlapping generations of young and old households. A young household has type $i = (y_i, a_i)$: income and wealth at entry, the wealth inclusive of what she inherited, net of the estate tax—newborns inherit the estates of the old who exit as they arrive. She decides whether to enter the housing market; inside, she chooses a tenure, a house size, completed fertility, and savings for old age. Old households, indexed by j , are of two kinds: old incumbents, last period's young owners, who enter old age with financial wealth, the house they bought, and a possible accrued gain, choose how much of the house to keep and how large an estate to leave, then exit; and old renters, last period's young renter entrants, who rent in old age and carry no wedge. Nonentrants stay outside the market in both periods. Each period a mass $\bar{M}$ of potential young households enters the pool: the maturing children of the previous cohort plus a renewal inflow. Types are distributed according to G_Y .

Preferences. Households have preferences over consumption, housing, and children. Children raise the household's consumption and housing needs: each child claims $\chi > 0$ units of consumption and $\kappa > 0$ units of space, so with consumption spending c_i , housing h_i , and fertility n_i ,

$$u_i^Y = \log(c_i - \chi n_i) + \alpha \log(h_i - \kappa n_i) + \vartheta \log n_i.$$

Old households value consumption, housing, and the estate $e_j \geq 0$ they leave at exit:

$$u^O = \log c_j^O + \gamma \log h_j^O + \mathcal{B}(e_j),$$

with $\mathcal{B}$ a warm-glow function, increasing and concave, possibly zero, common to all old households. The exiting cohort's estates are what the arriving cohort inherits. A young household also values her own old age: her objective adds βu^O , evaluated at the old-age choices she will make, with discount factor $\beta \in (0, 1]$.

Housing. There is a fixed stock $\bar{H}$ of divisible floorspace. Renting and owning are occupancy modes with different size menus,

$$h \leq \bar{h}^R \text{ (rent)}, \quad h \leq \bar{h}^O \text{ (own)}, \quad \bar{h}^R < \bar{h}^O,$$

Family-sized houses are mostly available to buy, not to rent. All floorspace trades in one market: the asset price is P , and no-arbitrage ties the rental rate to it,

$$q = \rho P, \quad \rho = r + \delta + \tau^p.$$

Given q , a higher property tax lowers P and with it the down payment on any given house. Lowercase h is an occupancy choice; capital H marks housing quantities not chosen today: the stock, the caps, carried housing.

Financing. A buyer finances at most a share $\phi \in (0, 1)$ of the purchase, so she posts a down payment $(1 - \phi)Ph$ at purchase, out of wealth in hand: income arrives only over the period, so a_i caps what she can put down. Lenders also apply a payment-to-income test: housing payments may not exceed the share $\psi \in (0, 1)$ of income. Wealth is otherwise fungible: it enters the budget alongside income, and buying assets is not expenditure, so the posted equity remains the household's. Households save: a young household carries $a'_i \geq 0$ into old age at the exogenous gross return $1 + r$, with no unsecured borrowing against old age; a renter, with no asset to pledge, therefore cannot borrow at all. The interest rate r is exogenous: households borrow and lend with a passive financial sector, so the only price determined inside the model is q .

Taxes. Three taxes appear: the property tax τ^p , part of the user cost above; the estate tax τ^b , collected when estates pass to the next cohort; and the capital-gains tax, the old-side institution at the center of the model. Selling a house triggers a tax: the seller pays the rate τ^g on the difference between the sale price and what she paid, in dollars, once that difference exceeds an exclusion. Dying instead passes the house with a stepped-up basis—the heir's purchase price is reset to the current value—so the gain is never taxed.

Where do gains come from when the real price never moves? From inflation. Dollar prices and incomes all grow at rate ϖ , so relative prices and real choices are unaffected; but the tax form compares dollars from different dates, and a house bought one period ago shows a paper gain even though its real value is unchanged. The tax on that paper gain is real,

$$\bar{\ell} = \tau^g \frac{P \varpi}{1 + \varpi}$$

per unit sold, and this is the model's only contact with inflation. An old incumbent's tax state is $\ell_j = \bar{\ell}$ if her gain clears the exclusion and zero if not; the exclusion is a dollar amount, so large holdings clear it and small ones do not. Everything is deterministic: a buyer knows the tax her future self will face.

2 Households

Conditional on entry, household i chooses a tenure mode $m \in \{R, O\}$, a bundle, and savings. For $m \in \{R, O\}$,

$$\begin{aligned} W_i^m = \max_{c_i, h_i, n_i, a'_i} & \log(c_i - \chi n_i) + \alpha \log(h_i - \kappa n_i) + \vartheta \log n_i \\ & + \beta [\mathbf{1}\{m = R\} V^R(a_{j'}) + \mathbf{1}\{m = O\} V^O(a_{j'}, h_i, \ell_{i'})] \end{aligned}$$

subject to

$$\begin{aligned} c_i + qh_i + a'_i &= y_i + a_i, & h_i &\leq \bar{h}^m, & a'_i &\geq 0, \\ \mathbf{1}\{m = O\} (1 - \phi)Ph_i &\leq a_i, & \mathbf{1}\{m = O\} qh_i &\leq \psi y_i. \end{aligned}$$

Rearranged, the budget prices a child:

$$(c_i - \chi n_i) + q(h_i - \kappa n_i) + (\chi + \kappa q)n_i + a'_i = y_i + a_i :$$

its consumption claim plus the market value of the space it occupies. A useful benchmark is equal scaling, $\chi = \kappa q / \alpha$, under which a child uses consumption and space in the same proportions as its parents.

The old state is generated by the household's own choices: financial wealth $a_{j'} = (1 + r)a'_i - \mathbf{1}\{m = O\}qh_i$ —real wealth is conserved, the nominal gain is not wealth—and the tax state $\ell_{i'} = \bar{\ell}$ if h_i clears the exclusion, zero below. Nothing is random; V^O and V^R are the old values defined next.

The old incumbent enters with financial wealth a_j , housing H_j^O , and the tax state ℓ_j . Each unit she holds faces one choice: sell now and pay the tax, or keep it until death, when the tax is forgiven. She solves

$$\begin{aligned} V^O(a_j, H_j^O, \ell_j) &= \max_{c_j^O, h_j^O, e_j} u^O(c_j^O, h_j^O, e_j) \\ \text{subject to } c_j^O + e_j + qh_j^O &= a_j + qH_j^O - \ell_j(H_j^O - h_j^O), & 0 \leq h_j^O &\leq H_j^O. \end{aligned}$$

Selling a unit nets $q - \ell_j$; the forgiveness at death is a subsidy to staying put, a transfer rather than a resource saving. An old renter has no carried housing and no tax:

$$V^R(a_j) = \max_{c_j^O, h_j^O, e_j} u^O(c_j^O, h_j^O, e_j) \quad \text{subject to } c_j^O + e_j + qh_j^O = a_j, \quad 0 < h_j^O \leq \bar{h}^R,$$

and at an interior choice her marginal value of space is exactly q .

Using $P = q/\rho$, owner housing is bounded by Using $P = q/\rho$, owner housing is bounded by

$$H_i^O = \min \left\{ \bar{h}^O, \frac{a_i \rho}{(1 - \phi)q}, \frac{\psi y_i}{q} \right\}, \quad H_i^R = \bar{h}^R.$$

H_i^m is the effective family-housing cap: a renter is capped by the stock, a buyer by whichever of menu, collateral, or debt service is shortest. Tenure is $W_i^Y = \max\{W_i^R, W_i^O\}$. Entry compares the inside value with an outside option $\bar{W}^E$ under extreme-value shocks of scale κ_E :

$$\pi_i^E = \frac{\exp\{W_i^Y / \kappa_E\}}{\exp\{W_i^Y / \kappa_E\} + \exp\{\bar{W}^E / \kappa_E\}}$$

is the probability that household i enters.

3 Equilibrium

Aggregate demand adds young households, old incumbents, and old renters, and the single market clears:

$$\bar{M} \int \pi_i^E h_i dG_Y(i) + \int h_j^O dF_O(j) + \int h_j^O dF_R(j) = \bar{H},$$

where F_O and F_R are the measures of old incumbents and old renters. A stationary equilibrium is a user cost q (and $P = q/\rho$) such that households optimize, the market clears, fiscal and credit accounts close with lump-sum transfers, next period's old households are this period's young entrants, owners as incumbents and renters as old renters, and births plus the renewal inflow keep the cohort at $\bar{M}$. Nothing in the clearing condition asks whether the marginal family-sized unit is held by the household that values it most; that is the next section's question.

4 Equilibrium policies

How much a constraint binds can be read off the household's own bundle. The case problem's Lagrangian is

$$\begin{aligned} \mathcal{L} = & u_i^Y + \beta V_{j'} + \lambda_i (y_i + a_i - c_i - qh_i - a'_i) \\ & + \mu_i (a_i - (1 - \phi)Ph_i) + \eta_i (\psi y_i - qh_i) + \theta_i (\bar{h}^m - h_i) + \xi_i a'_i, \end{aligned}$$

with $\mu_i = \eta_i = 0$ in the rental mode and $\xi_i \geq 0$ the multiplier on the no-borrowing bound $a'_i \geq 0$ —for a renter, the constraint that she cannot borrow at all; every condition in this section is a derivative of $\mathcal{L}$.

The first-order condition for saving sets the marginal value of resources today against the return tomorrow:

$$\frac{1}{c_i - \chi n_i} = \beta(1 + r) \frac{1}{c_{j'}^O} + \xi_i, \quad \xi_i a'_i = 0,$$

which for savers ($\xi_i = 0$), under logarithmic old utility (warm glow $b \log e$ included) and interior old choices, collapses to a savings rule:

$$a'_i = k(c_i - \chi n_i) + \frac{\ell_{i'} h_i}{1 + r}, \quad k = \beta(1 + \gamma + b) :$$

the household saves old age's claim on adult consumption plus her discounted future tax bill, so anticipating the lock-in tax is itself a savings motive. By the same conversion, a buyer above the exclusion pays an effective price $q^O = q + \bar{\ell}/(1 + r)$ per unit of space—the market price plus her own discounted lock-in tax—while a renter or a below-exclusion buyer pays q ; write q^m for the case's effective price. A unit of adult consumption commits $1 + k$ units of resources, one spent now and k carried.

When no cap binds, the allocation follows in one line: with lifetime resources $w_i = y_i + a_i$,

$$c_i - \chi n_i = \frac{w_i}{1 + \alpha + \vartheta + k}, \quad n_i = \frac{\vartheta w_i}{(1 + \alpha + \vartheta + k)(\chi + \kappa q^m)}, \quad h_i = \frac{\alpha w_i}{(1 + \alpha + \vartheta + k) q^m} + \kappa n_i,$$

and savings follow the rule. When the cap binds, $h_i = H_i^m$, fertility solves the condition displayed in the last section, and the budget gives the rest.

Dividing the housing derivative of $\mathcal{L}$ by the consumption derivative, $\lambda_i = 1/(c_i - \chi n_i)$, prices every multiplier in goods. For a renter this gives

$$\frac{\alpha(c_i^R - \chi n_i^R)}{h_i^R - \kappa n_i^R} = q + \zeta_i^R, \quad \zeta_i^R = \frac{\theta_i}{\lambda_i} \geq 0 :$$

the left side is her marginal value of space—the consumption she would give up for one more unit—and it exceeds the rent exactly by the bite of the rental menu, computable from her observed bundle. For a young owner the same two derivatives give

$$\frac{\alpha(c_i^O - \chi n_i^O)}{h_i^O - \kappa n_i^O} = q^O + \zeta_i^O, \quad \zeta_i^O = \underbrace{\frac{\mu_i}{\lambda_i}(1 - \phi)P + \frac{\eta_i}{\lambda_i}q}_{\zeta_i^{O,F}, \text{ financing}} + \underbrace{\frac{\theta_i}{\lambda_i}}_{\text{menu}} :$$

her marginal value of space exceeds her effective price by the bite of whichever access constraints bind—the down payment (μ_i), the payment-to-income test (η_i), the owner menu (θ_i). When the menu is slack, the bundle identifies the financing bite alone: $\zeta_i^{O,F}$ is the observed ratio minus q^O . For an old incumbent at an interior retention choice,

$$\frac{\gamma c_j^O}{h_j^O} = q - \ell_j :$$

she keeps marginal space she values below what the market would pay, because selling it would realize the taxable gain. Old renters carry no such wedge: their marginal value is q .

The gaps price children. Multiplying the fertility first-order condition by consumption gives, in either tenure,

$$\frac{\vartheta(c_i^m - \chi n_i^m)}{n_i^m} = \chi + \kappa(q^m + \zeta_i^m),$$

with q^m the case's effective price. A child's space requirement is priced at the household's own shadow value of space. Children get more expensive exactly where family-sized space is hard to reach, with market prices unchanged.

5 Efficiency

The planner faces the economy's real constraints and nothing else: the stock, the size menus, goods feasibility, and the domain conditions. It does not face household budgets, since allocations are supported by lump-sum transfers, and it does not face the financing inequalities, which restrict market purchases rather than occupancy. Let $r_i^F \geq 0$ be the marginal real cost of implementing one more unit of owner space for a financing-constrained household: zero if the planner can reassign directly, positive if it must work through costly instruments, infinite if the constraint is a primitive nothing relaxes.

Proposition 1. *Hold the stationary cross-section, entry, tenure assignments, and savings fixed. The planner's surplus from moving one unit of space from an old incumbent at an interior retention choice to a young owner whose owner menu is slack and whose financing constraints bind is*

$$(q^O + \zeta_i^{O,F} - r_i^F) - (q - \ell_j) = \zeta_i^{O,F} + \frac{\ell_{i'}}{1+r} + \ell_j - r_i^F.$$

The competitive allocation solves the planner's conditional problem only if no feasible pair has positive surplus. Conversely, if all financing gaps and levies are zero, the allocation satisfies the planner's conditional first-order conditions.

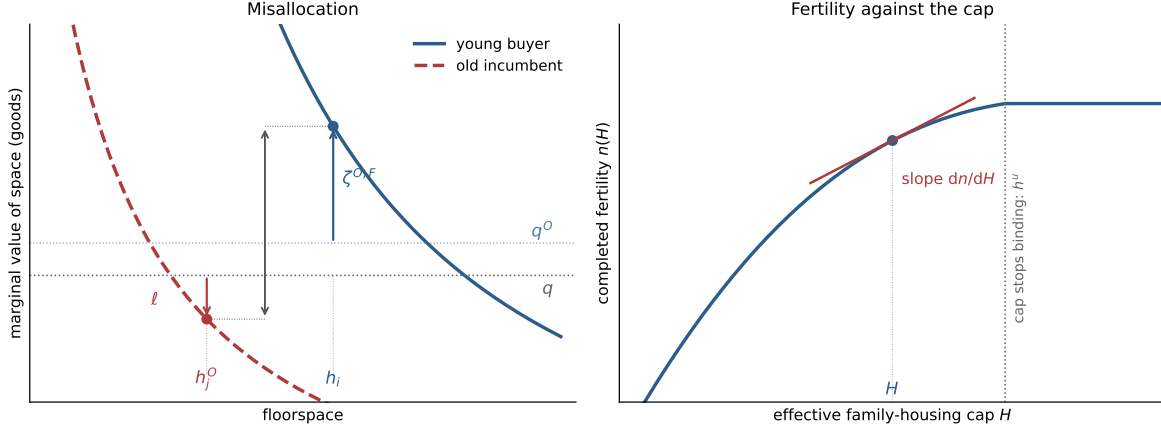


Figure 1: Misallocation and the fertility margin. Left: the collateral-capped buyer values marginal space at $q^O + \zeta^{O,F}$, above the market price; the locked-in incumbent values it at $q - \ell$, below; the distance between the two points is the recoverable surplus. Right: completed fertility against the effective family-housing cap, with slope dn/dH at the equilibrium cap; the cap stops binding at h^u .

To prove it, move dh from the incumbent to the buyer and hold both at their prior utility with lump-sum transfers; the residual $(\zeta_i^{O,F} + \ell_i/(1+r) + \ell_j - r_i^F)dh$ in goods is left over, a Pareto improvement under any welfare weights; both tax terms are transfers the planner recovers. Two cautions. Real menus are not market failures: a planner facing the same stock is pinned with the renter at $\bar{h}^R$. And a positive financing gap is an inefficiency only when an instrument can reach it at real cost below the value gap.

The two margins in one picture. Figure 1 draws a solved stationary case as theory. On the left, the buyer's marginal value of space falls along her constrained path and sits above her effective price by the financing gap $\zeta^{O,F}$ at her cap, while the incumbent's sits below the market price by her tax ℓ ; moving one unit between them frees $\zeta^{O,F} + \bar{\ell}/(1+r) + \ell$ in goods. On the right, completed fertility rises with the cap at rate dn/dH until the cap stops binding at h^u .

6 Fertility and policy

Fix a tenure case, let H be the effective cap, write $w = y + a$ for lifetime resources, and let q^m be the case's effective price. Concentrating the savings rule into the budget, a binding cap makes fertility solve $\vartheta/n = (1+k)\chi/(w - q^m H - \chi n) + \alpha\kappa/(H - \kappa n)$, and

$$\frac{dn}{dH} = \frac{\frac{\alpha\kappa}{(H - \kappa n)^2} - \frac{(1+k)\chi q^m}{(w - q^m H - \chi n)^2}}{\frac{\vartheta}{n^2} + \frac{(1+k)\chi^2}{(w - q^m H - \chi n)^2} + \frac{\alpha\kappa^2}{(H - \kappa n)^2}},$$

which is positive for every binding cap if and only if $\chi \leq (1+k)\kappa q^m/\alpha$: relaxing access raises fertility whenever the child's consumption requirement does not exceed its space requirement priced at the lifetime cost of adult consumption. The savings margin loosens the condition because forced housing spending now crowds out saving alongside consumption.

At fixed prices, a small policy z moves aggregate fertility through the constrained:

$$\left. \frac{dN}{dz} \right|_q = \bar{M} \sum_m \int_{\mathcal{E}_m(z)} \left[\pi_i^E \Psi_i^m + n_i^m \frac{\pi_i^E (1 - \pi_i^E)}{\kappa_E} \Theta_i^m \right] \mathcal{P}_i^m(z) dG_Y(i),$$

where $\mathcal{E}_m(z)$ is the set of households whose cap binds in case m , $\mathcal{P}_i^m = \partial H_i^m / \partial z$ is the pass-through of the policy to the binding component of the cap, $\Psi_i^m = dn/dH$ is the fertility response, and $\Theta_i^m = \zeta_i^m / (c_i^m - \chi n_i^m)$ drives the entry term. A policy aimed at a slack constraint moves nothing: a down-payment grant does nothing for a household pinned by the rental menu. A transfer of spendable resources also moves fertility through the budget at a fixed cap; the formula isolates the access channel. At $\chi = (1 + k)\kappa q^m / \alpha$, equal scaling at lifetime prices, Ψ_i^m is proportional to $\zeta_i^m / (c_i^m - \chi n_i^m)$, so the same gap that measures misallocation predicts whose fertility responds.

Because tenure is a discrete choice, a policy that shifts $W_i^O - W_i^R$ also moves households across the tenure boundary. Below the exclusion both tenures price space at the same q , an indifferent household has the same allocation in both, and the switching jump is exactly zero. Above it, the anticipated tax makes owning the dear way to occupy space, and it also hands the owner case a constant old-age subsidy—occupying retained space at $q - \bar{\ell}$ —that shifts who sits at the boundary without moving any bundle. The long version prices the tension: the switcher into ownership upsizes when the wedge dominates the subsidy in resources, and she has more children when the space force dominates the subsidy’s level shift; at the example’s parameters both hold with room, and the boundary household upsizes from 0.15 to 0.20 and adds 0.007 children. The long version of the note states and proves this, together with the stationary demographic closure and the full equilibrium definition.